

Department of Computer Science**Assignment 1****Course Code: CS-112****Name:** _____**Course Title: Object Oriented Programming****Registration No.** _____**Total Marks: 20****Discipline/Semester: BS(CS) 2nd A/B****Assigned Date: 22-03-2025****Due Date: 25-03-2025**

Instructions:

- Late submissions will not be entertained.
- Copied assignments will be marked ZERO.
- Submit the assignment in hard form
- Use of AI tool is strictly not allowed.

CLO1, GA2, C2

Question 1:

Discuss compile time and runtime errors and Explain these with respect to the given scenario. How would you ensure that these errors do not occur.

Scenario: An airline company wants to create a simple flight booking system where passengers can book a seat on a flight if seats are available. Each flight has a flight number, destination, and a limited number of seats.

Tasks:

1. Create a Flight class with attributes:
 - flightNumber
 - destination
 - totalSeats
 - availableSeats
2. Implement a method bookSeat() that checks if a seat is available:
 - If available, decrease availableSeats by 1 and print "Booking successful!"
 - If no seats are available, print "Sorry, no seats available."
3. Implement a method displayDetails() to show the flight number, destination, and available seats.
4. In the main method, create an array of Flight objects for different flights.
5. Allow the user to select a flight by entering its flight number and book a seat.